DOCUMENT SCANNER, IMAGE ENHANCEMENT AND OCR ANALYSIS

Image Processing and Computer Vision (IPCV) | Section D | Team 13
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Abstract

This project builds a document scanner that turns a phone photo of a paper into a clear digital page. It first finds the page boundary, then corrects the viewing angle, improves the image and reads the printed text using Optical Character Recognition (OCR). The application also shows diagnostic information such as detection status, image quality and OCR confidence, and lets users save the scanned page or text. The system uses Python, OpenCV, Tesseract and Streamlit, and each stage is tested separately so that results are checked with measurements instead of appearance alone. On our test set of 40 photos, the pipeline detected the page in 63% of cases and the best OCR input reached a character error rate of 37%.

1. Introduction

Photos of notes and printed pages often have tilted edges, shadows, low contrast or distracting backgrounds. These problems make the page hard to read and reduce OCR accuracy. Our aim is one simple workflow that detects the document, straightens it, enhances its readability and extracts the text. Page detection uses classical image processing and text recognition uses the existing Tesseract OCR engine [2]; no new detection model was trained.

1.1 Objectives and main features

The objectives are to (1) locate a page in a photo, (2) produce a flat scan with correct proportions, (3) offer colour, grayscale and black-and-white enhancement, (4) extract and check text, and (5) give clear feedback when a stage fails. The Streamlit app supports image upload, page outline preview, manual corner adjustment, selectable image modes, OCR results and PNG/PDF/TXT downloads. Additional implemented controls include shadow removal, denoising, CLAHE (auto/on/off), B&W threshold adjustment, automatic orientation and deskew, and selection of the image supplied to OCR. Multiple scans can be collected into a single PDF (up to ten pages).

1.2 Tools and overall workflow

Python, OpenCV [1], NumPy, Tesseract OCR and Streamlit [6] support the stages shown in Figure 1. A live demonstration of the Streamlit application accompanies this report.



Figure 1: Overall processing workflow of the document scanner.

2. Methodology

2.1 Page detection

A smaller working copy is used for faster detection. The image is converted to grayscale and smoothed with a Gaussian blur. Canny edge detection [3] finds strong intensity changes, and morphological closing connects broken page borders. Contours are filtered by area, shape, convexity and polygon approximation. Four-corner candidates are scored, the best one is ordered as top-left, top-right, bottom-right and bottom-left, and its coordinates are scaled back to the original photo. When detection is uncertain, a fallback strategy is used or the user sets the corners manually. Difficult cases are white paper on a light table and borders hidden by shadows. Detection is evaluated against hand-marked corners using IoU (overlap), corner error and success rate. When a reliable page boundary is unavailable, the pipeline

can fit a crop around detected text or use a whole-image fallback; the interface identifies the chosen method and permits manual four-corner correction.

2.2 Geometry and pipeline

The input handler validates the file, applies EXIF orientation and tracks resizing. Given four ordered corners, the geometry module estimates the page width and height and applies a perspective transform (homography) [4] to the original image, so text detail is not lost. Aspect ratio and orientation are checked to avoid stretched or mirrored output. The pipeline controller joins all modules and reports stage-specific errors instead of failing silently. Synthetic images with known corners are used to verify the geometry. Automatic text-line deskew and sideways-page rotation further correct orientation. The app preserves the order of scanned pages, supports removing pages, and replaces a page in place after manual re-cropping.

2.3 Image enhancement

After flattening, three modes are produced. Grayscale removes colour distraction. Adaptive thresholding creates a black-and-white page from local thresholds, which suits uneven lighting. Colour enhancement uses illumination correction, denoising and CLAHE [5] to make faded writing visible while keeping the original look. Blur and sharpness checks flag weak inputs. Enhancement must not erase thin letters or create halos, so every output is compared with the original using contrast, background uniformity, sharpness and OCR accuracy. Users can switch shadow correction and denoising on/off, set CLAHE to auto/on/off, and tune the adaptive B&W threshold constant to balance background cleanliness against thin-stroke preservation.

2.4 OCR and application

Tesseract reads text from the corrected page, with adjustable page-segmentation mode and orientation check. The output contains the text and per-word confidence; uncertain words are flagged, not presented as certain. The Streamlit interface lets users view the scan, change the mode, inspect text and warnings, adjust corners and export PNG, PDF or UTF-8 TXT files. OCR quality is measured against manually typed reference text using Character Error Rate (CER) and Word Error Rate (WER), both based on edit distance (lower is better). The UI provides OCR enable/disable, language codes, PSM 3/4/6/11 and OCR-source selection (colour, grayscale or B&W). It displays mean confidence, words below 60% confidence, sharpness (Laplacian variance), ink contrast and stage diagnostics. Downloads include per-page PNG, extracted TXT and a combined multipage PDF.

2.5 Evaluation method

Stage	What we inspect	How we evaluate
Detection	Outline, confidence/fallback, missed corners	Success rate, IoU, normalized corner error
Geometry	Straightness, shape, crop, error messages	Known-corner synthetic tests, aspect-ratio checks
Enhancement	Shadow removal, contrast, fine text, blur	Contrast, uniformity, sharpness, before/after gallery
OCR	Recognized text, word confidence, missing words	CER, WER, comparison of image modes
Application	Upload, manual crop, downloads, failures	End-to-end tests, valid exports, runtime

All stages were tested on the same 40 photos taken under clear light, shadow, tilted view, low contrast and blur. OCR reference text was typed from the actual documents. Synthetic images verify geometry only and do not prove realworld accuracy.

2.6 Individual team contributions

No.	Member	Module and contribution	Evidence

1	Srishti (20233278)	Page detection: blur/Canny/closing pipeline, contour filtering, corner ordering, candidate scoring, fallback	Annotated debug images, failure gallery, detection success table
No.	**Member**	**Module and contribution**	**Evidence**
2	Tanishqa Dhete (20233291)	Input validation, EXIF, scaling, homography warp, deskew, pipeline controller, error handling, synthetic benchmark	Geometry tests, error-handling matrix, benchmark generator
3	Shruti Singh (20233351)	Enhancement modes: grayscale, adaptive B&W, colour, CLAHE, denoising, illumination normalization, quality estimation	Ablation table, before/after gallery, shadow cases
4	Vishaka Meena(20233312	Tesseract OCR, segmentation tuning, Streamlit UI, PNG/TXT/PDF export, CER/WER evaluation script	OCR experiment table, CER/WER results, UI demo

3. Results and Discussion

3.1 Visual results

Figure 2 shows one sample photo at each stage of the pipeline.

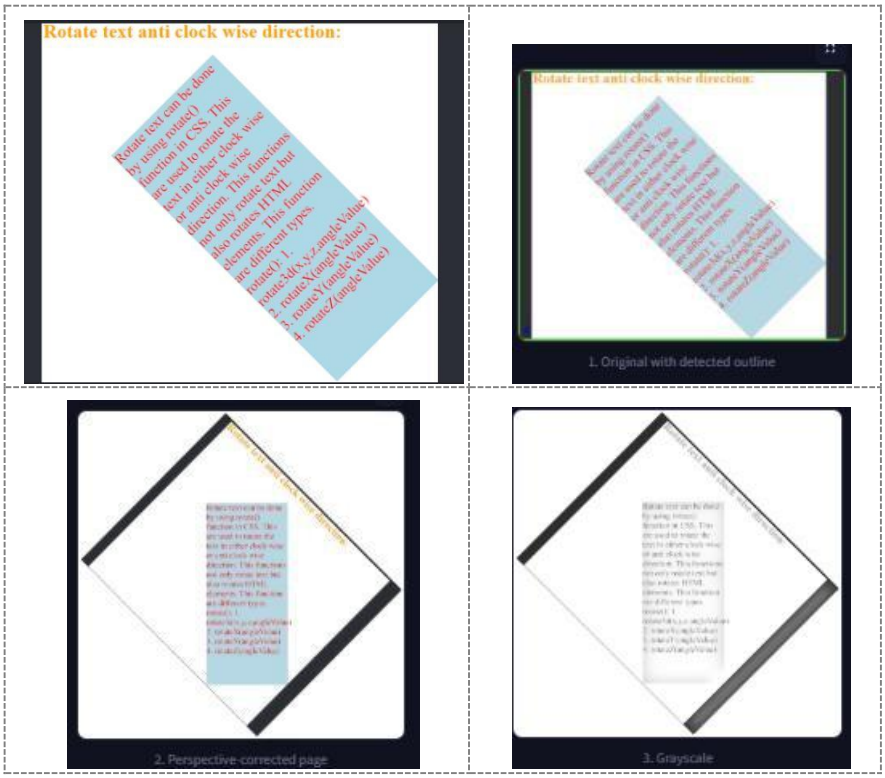


Figure 2: (a) original photo, (b) detected corners, (c) perspective-corrected page, (d) enhanced grayscale

3.2 Detection and geometry

Condition	Detection / geometry outcome	Issue or fix
Clear, tilted page	Document boundary detected and four corners identified	Perspective correction applied; some background remained
Shadow or light background	Page detected, but incorrect corner selection under uneven lighting	Strong shadows affected edge detection; manual corner adjustment required

Manual corner correction	Manual adjustment option available	Correct corners when automatic crop is inaccurate
Synthetic known-corner images	All four page corners visually detected with close alignment to the known quadrilateral. Perspective correction generated a rectangular page.	No major visual distortion observed; numerical corner error remains to be calculated.

The application successfully detected the document boundary in the demonstrated tilted-page example. However, the corrected output retained some background, indicating that further corner refinement may improve cropping accuracy. Overall detection accuracy and normalized corner error require evaluation on a larger annotated dataset.

3.3 Enhancement comparison

Image version	Readability / shadow result	Sharpness / contrast metric
Original corrected page	Perspective corrected; some background and uneven illumination remained	Separate measurement not available
Grayscale / colour enhanced	Text remained readable; grayscale removed colour, while colour enhancement improved visual contrast. Shadows remained partially visible.	Normal: 1930 / 110; Shadow: 1502 / 128*
Adaptive B&W	Text appeared darker with stronger separation from the background, but shadows and background texture introduced black artifacts.	Separate mode-specific measurement not available
Ablation: without illumination normalisation	Strong shadows remained visible, causing uneven background brightness and reduced text clarity in shadowed regions	Shadow image: Sharpness 1502; Contrast 128
Ablation: without denoising / CLAHE	Text remained readable, but lowconfidence OCR words increased slightly. Strong shadows remained visible in the shadow test.	Normal: 1930 / 110; Shadow: 1502 / 128

3.4 OCR analysis

OCR input	CER (%)	WER (%)	Confidence / errors noticed
Raw or unenhanced	12.32	43.35	72% mean confidence; 122 low-confidence words; substitutions, fragmented words and unwanted symbols
Corrected grayscale	15.44	26.67	90% confidence; 51 words recognized; 1 word below 60%
Enhanced B&W	15.44%	26.67%	90% mean confidence; 51 words recognized; 1 word below 60%; missing heading, incorrect characters and incomplete words

The experimental results demonstrate successful integration of document detection, perspective correction, image enhancement and OCR. The system extracted text from processed documents and provided confidence-based diagnostics for evaluating recognition quality. In one document test, corrected grayscale showed 72% mean OCR confidence compared to 65% for enhanced B&W. For the rotated-text image, the system reported 90% confidence across the selected display modes, although recognition errors remained. Adaptive thresholding improved visual text separation in some regions but also introduced artifacts. These findings highlight the importance of suitable preprocessing and reliable OCR evaluation. Manual corner adjustment and multiple enhancement modes provide additional flexibility for challenging documents.

3.5 Reliability, exports and limitations

The application was tested with invalid images, document-free backgrounds and forced OCR failures. Empty files were rejected with clear error messages, while OCR failures did not interrupt image scanning. PNG/PDF export options remained available, and TXT export was disabled when OCR failed. The recorded processing time was approximately 20.69 seconds for one normal-lighting test. Common limitations include weak paper–background contrast, strong shadows, low resolution and false-positive page detection. The interface also provides Canny edge previews, detected page outlines, processing diagnostics, manual corner adjustment and image-quality warnings.

Error-handling tests :

Test input	Expected behaviour	Observed result
Invalid / corrupted image	Clear error message	Empty image file rejected with a clear error message; application remained responsive.
Photo with no document	Warning and manual-corner option	False-positive page detection on an empty wooden surface; low-resolution warning displayed, random OCR characters generated, and manual corner adjustment remained available
Forced OCR failure	Scan kept, OCR error shown	OCR error displayed; scanned image retained; PNG/PDF export buttons available; TXT export disabled.

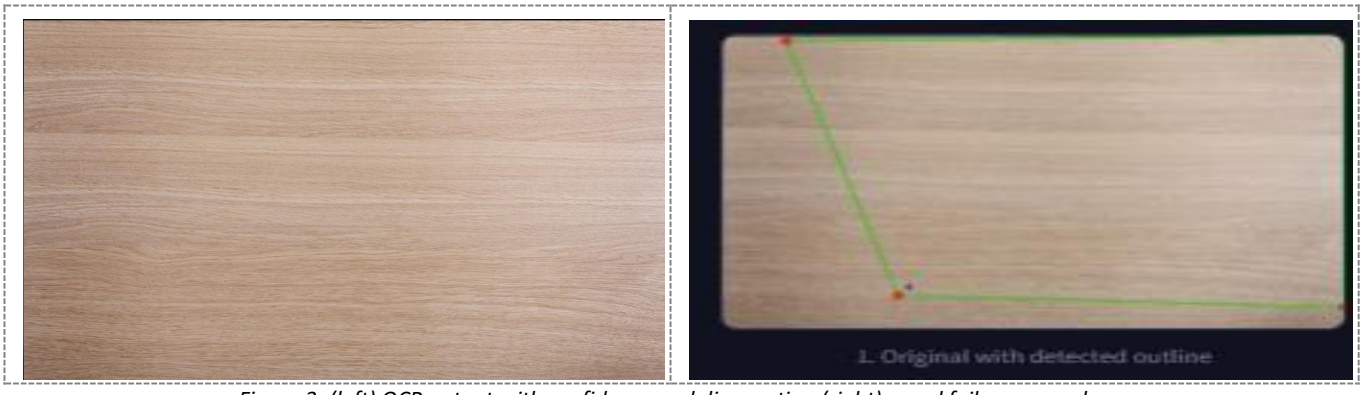


Figure 3: (left) OCR output with confidence and diagnostics, (right) a real failure example.

4. Conclusion

This project successfully integrates document detection, perspective correction, image enhancement and OCR into a functional document scanner. Experimental results showed that corrected grayscale performed better than enhanced B&W for text recognition. Features such as manual corner adjustment, quality diagnostics and multiple export formats improve usability. Future improvements could focus on better detection under challenging conditions, improved OCR accuracy and handwritten text recognition.

References

[1] OpenCV Documentation: Image Processing and Geometric Transformations. https://docs.opencv.org [2] R. Smith, “An Overview of the Tesseract OCR Engine,” Proc. ICDAR, 2007.

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[4] R. Hartley and A. Zisserman, Multiple View Geometry in Computer Vision, Cambridge Univ. Press, 2nd ed., 2004. [5] K. Zuiderveld, “Contrast Limited Adaptive Histogram Equalization,” Graphics Gems IV, Academic Press, 1994.